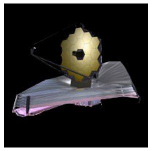

6241 - Characterizing Supernova Progenitor Stars in Archival JWST Imaging

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Supernova progenitor star research has greatly benefitted from vast NASA archives in which individual massive star systems can be resolved in pre-explosion imaging of nearby (<40 Mpc) galaxies. Inevitably, there will be supernova explosions in available JWST imaging, requiring careful analysis of NIRCam and MIRI data in order to isolate, measure photometry, and characterize the type of star that exploded. This analysis will provide new insight into the heavily reddened progenitor systems of Type II supernovae similar to the recent SN2023ixf in M101, stripped-envelope supernova progenitor stars in highly extinguished and crowded environments, the eruptive and enigmatic progenitor stars of Type IIn supernovae, and new discoveries we have not yet been able to probe due to a lack of deep, high-resolution, mid-infrared data. New discoveries are already occurring, including the pre-explosion counterpart to the Type Ibn SN2023fyq in the dusty, inclined spiral galaxy NGC 4388 at ~20 Mpc, whose analysis this program will directly support. The final products from this program will be photometry and deep upper limits on all new supernova progenitor star discoveries with archival JWST imaging as well as the imaging analysis tools, models, and software necessary to analyze their optical to mid-infrared spectral energy distributions.

OBSERVING DESCRIPTION

This archival research proposal aims to analyze 155 NIRCam and MIRI data sets covering 12 galaxies within 25 Mpc in order to find red supergiants in their stellar populations. These data comprise all available medium- and wide-band NIRCam and MIRI filters within 1 arcminute of the target galaxies, totaling anywhere from 6-29 filters per galaxy but always with a single "optical" (<1.5 micron; either F090W or F115W), "near-infrared" (F150W), and "mid-infrared" (F444W) filter in order to optimally cover red supergiant spectral energy distributions. We will analyze all such data, use them to isolate red supergiants in color-magnitude space, and analyze the distributions of these stars as a function of their basic physical properties and local environment in order to achieve our overall science goals.